Course Name:	**Artificial Intelligence Machine Learning - Deep Learning**
Course Duration:	3 Months
No. Of Days per week:	4 Days per week (Course only)
Class Hours	3 hours per class
Total In-Class Credit hours	144

Module	Week	Content Breakdown	Tasks
1. Foundation	1,2	Unit 1: Mathematics for AI/ML: Linear algebra (vectors, matrices), calculus (derivatives, gradients), basic optimization.	Task 1 Task 2 Task 3-25 Task 26- 27 Task 49- 51
		Unit 2:Probability & Statistics: Probability theory, Bayes' theorem, descriptive statistics, inference.	
		Unit 3: Programming Foundations: Core Python (syntax, functions, loops), libraries (NumPy, Pandas, Matplotlib, Scikit-learn).	
		Quiz and Tasks	
		Assignment: Write Python scripts for basic data analysis. - Simulate probability concepts and visualize math operations using NumPy. - Explore AI applications through case studies.	
2. AI/ML and Cloud Platforms	3	Unit 1:AI Basics and Real-world Applications: Use cases of AI and ML in industries such as healthcare, finance, e-commerce, and transportation.	Task 28- 48
		Unit 2:AI pipeline	
		Unit 3: AI Cloud Ecosystem - Cloud: Core ML Services - Cloud: Development Tools	
		Quiz and Tasks	
		Assignment: Setting up cloud Envionment and related tools	
3. Data Pipeline: Data Preparation and Feature Engineering - Data Analysis	4	Unit 1: Data Preprocessing: Data collection, cleaning, transformation, encoding, scaling.	Task 28-48
		Unit 2: Data Engineering with cloud	
		Unit 3: Data Pipelines: ETL processing, stream and batch handling, automation.	
		Quiz and Tasks	
		Assignment (1.5 hr per week)**:** - Build an end-to-end data pipeline using BigQuery ML and Dataflow. - Perform data wrangling and transformation using cloud tools.	
4. Basic AI and ML	5,6	Unit 1: Supervised Learning: Regression (linear/logistic), classification (SVM, KNN, naive Bayes).	Task 51,52 Task 53- 55
		Unit 2: Decision trees, and Ensemble models	
		Unit 3: Perceptron Learning, and Neural Networks	
		Unit 4: Model Selection: Bias-variance tradeoff, evaluation metrics, confusion matrix.	
		Quiz and Tasks	
		Assignment (1.5 hr per week)**:** Build and train models. - Tune hyperparameters - Compare model types on a small dataset.	
5. Advanced AI and ML	6,7	Unit 1: Deep Learning Models: CNNs, RNNs, transformers, attention models.	Task 56-64
		Unit 2:Optimization Techniques: Regularization, dropout, learning rate tuning.	
		Unit 3: Cloud-Based Training: TPU/GPU distribution on GCP.	
		Quiz and Tasks	
		Assignment: - Fine-tune pre-trained models. - Perform transfer learning. - Run distributed training.	
6. Unsupervised Learning	8	Unit 1: Clustering K-Means, DBScan etc.	Task 56-64
		Unit 2: Dimensional Reduction	
		Quiz and Tasks	
		Assignment: - Define, develop and analyse unsupervised models.	
8. ML Pipeline and Automation	9	Unit 1: MLOps, Principles, Lifecycle, Architecture	
		Unit 2: Design and Development	
		Unit 3: MLOps Best Practices: CI/CD, versioning, dataset/model tracking	
		Unit 4: Deployment	
		Unit 5: ML Workflow Automation: Kubeflow Pipelines, Cloud Composer.	
		Quiz and Tasks	
		Assignment: Hands-On: - Deploy and automate ML pipelines for real-time predictions.	

9. Model Optimization, Performance Tuning, Deployment	10	Unit 1: Model Optimization: Pruning, quantization, knowledge distillation. Unit 2:Performance Monitoring: Model drift, accuracy, retraining strategies. **Quiz and Tasks** **Assignment:** - Optimize an image recognition model using TensorRT. - Monitor and retrain model using Vertex AI tools.	Task 65
11. Responsible and Explainable AI	11	Unit 1: Bias Detection: Fairness indicators, model evaluation across demographics. Unit 2:Explainability: SHAP, LIME, What-If Tool (WIT), Physics Aware AI. Unit 3: Ethical AI: Google frameworks for responsible ML. **Quiz and Tasks** **Assignment:** - Evaluate model bias. - Apply interpretability tools to explain predictions.	Task 65
12. Generative AI	12	Basics, Technologies and Applications	Task 65
13. Capstone Project	9,10,11,12	Project Runs Parrallel focusing on Real-World AI Applications: Design, build, and deploy an AI/ML solution. Project Examples: Medical diagnostics, fraud detection, recommender system, NLP etc. Hands-On: - Build an end-to-end ML pipeline on cloud. - Present and evaluate the project outcome	1. Develop, discuss and improve project 2. Project Presentations

Notes:

Activities & Tasks	Tasks as per NAVTTC Course Annexure-I, however Quiz, assignments and Capstone Project are additional
Motivational Lecture	We have a separate Soft Skill program which will be conducted every week on Soft Skill day. An Outlined draft of our Soft Skills program is attached Seperately

Item	**NIAI**	**Remarks**
Course Duration	3 Months	All NAVTTC topics covered + Corporate development Soft Skills Program
Total Class Days	5 Days	4 days/week (Course) + 1day for soft skills.
In-Class Hours	3 hours/day	Content coverage sustained
Total In-Class Hours	144 hours (Excluding notional hours)	In-Class subject: 144 hours In-Class Soft Skills: 24 hours Project work: Approx. 100 hours (guided independent and team-based projects)

Soft Skills & Leadership Modules

Delivered by NIAI's (NETSOL Institute of Artificial Intelligence) Organizational Development experts, these modules enhance corporate readiness and professional competencies:

- **Building Future Ready Tech Leaders**
- **Communication Improvement Program**
- **Teamwork & Collaboration**
- **Emotional Intelligence & Empathy**

This stream adds 2 hours/week across **12 weeks (24 hours),** reinforcing Professionalism and Business Skills competencies vital for workplace integration.